

Inference Engineering Week Task

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Part A: Theory

1. What is the Difference Between Latency and Throughput in Model Inference?

Latency and throughput are both used in inference engineering to measure the performance of LLMs.

Latency: Latency is the time required by a model to process a single task/input and produce the output result.

Throughput: Throughput is the number of requests or samples that a model can process in a given amount of time.

Which factor influences model performance more?

It depends on the application. For real-time applications such as chatbots, low latency is more important because users expect a quick response. For applications that process many requests at the same time, high throughput is more important because more requests can be handled in a given amount of time.

2. KV Cache

KV Cache (Key-Value Cache) is a technique used in Transformer-based language models to speed up text generation by avoiding the same computation for tokens.

The model generates one token at a time. For every token, the attention mechanism needs the Key and Value representations of previous tokens.

Instead of calculating these values again for every new token, the model stores previously calculated K and V values in memory and reuses them.

Example:

If a model generates “THIS IS EXAMPLE OF KV CACHE” and wants to generate the next token, the K and V values for “THIS IS EXAMPLE OF KV CACHE” can be reused instead of recomputing them.

Trade-off: KV Cache requires additional memory.

3. What is Quantization?

Quantization reduces the numerical precision used to store or compute model weights.

For example, model weights can be converted from 32-bit or 16-bit floating-point numbers to 8-bit or 4-bit integers. This reduces model size and can speed up computation.

Examples:

- FP32 $\rightarrow$ 32-bit floating point
- FP16 $\rightarrow$ 16-bit floating point
- INT8 $\rightarrow$ 8-bit integer
- INT4 $\rightarrow$ 4-bit integer

Quantization Formula:

$$\text{scale} = \frac{x_{\max} - x_{\min}}{q_{\max} - q_{\min}} \quad (1)$$

$$\text{zero_point} = q_{\min} - \frac{x_{\min}}{\text{scale}} \quad (2)$$

4. Real Inference Serving Framework

vLLM is an inference and serving framework designed for large language models. It is used to efficiently serve LLMs and handle multiple requests.

One important optimization used by vLLM is **PagedAttention**.

PagedAttention manages the KV cache in smaller blocks or pages instead of requiring one large contiguous memory allocation for each sequence.

Example:

If many users send requests to an LLM at the same time, KV Cache allows vLLM to serve more requests without wasting large portions of memory.